PUC-SP • MAPPING MACHINE LEARNING DECISIONS

Class 12: Threshold of Decision and Calibration

From the Chosen Model to the Reliable Decision in Production

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Traditional accuracy fails under class imbalance and asymmetric error costs

Default 0.50 Cutoff Fallacy

- Assuming a 0.50 cutoff assumes symmetric error costs and a 50/50 balanced dataset.
- In clinical triage (15% prevalence), 0.50 misses >52% of sick patients.
- A naive model predicting all negative gets 85% accuracy but zero diagnostic utility.

Asymmetric Error Costs

- False Positive (FP): Healthy patient held for extra test (low operational cost).
- False Negative (FN): Sick patient sent home untreated (high critical/fatal cost).
- Principle: Cost(FN) >> Cost(FP). Optimization must reflect business impact.

Dual Pillars of Reliability

1. Ranking Discrimination: Capacity to separate positive from negative cases (measured via ROC-AUC).
2. Probability Honesty: Alignment of predicted probabilities with real observed rates (measured via Reliability Diagrams).

Logistic Regression won the benchmark due to strong ranking capacity and superior stability

Algorithm	ROC-AUC	Std Dev (σ)	Status
Logistic Regression	0.960	±0.014	SELECTED (Winner)
Random Forest (200 Trees)	0.945	±0.025	Statistical Tie
K-Nearest Neighbors (K=15)	0.940	±0.021	Eliminated
Decision Tree	0.803	±0.038	Eliminated

Variance Tie-Breaking Rule

- Statistical Overlap:
  Logistic Regression (0.960 - 0.014 = 0.946) overlaps with Random Forest (0.945 + 0.025 = 0.970).

- Decision Rule:
  When mean scores overlap within confidence intervals, select the architecture with lower variance (σ = 0.014 vs σ = 0.025).

- Why ROC-AUC?
  Evaluates ranking across ALL possible thresholds without arbitrary 0.50 commitment.

Reducing decision threshold to τ = 0.14 cut total operational error cost by 44%

Operational Cost Function: Total Cost = (1.0 × False Positives) + (5.0 × False Negatives)

Default Cutoff (τ = 0.50)

- False Positives (FP): 7 cases (Cost = 7)
- False Negatives (FN): 35 cases (Cost = 175)

- TOTAL OPERATIONAL COST = 182 units

- Impact: High accuracy on paper, but misses 35 high-risk sick patients due to overly strict threshold.

Optimal Cost Cutoff (τ = 0.14)

- False Positives (FP): 67 cases (Cost = 67)
- False Negatives (FN): 7 cases (Cost = 35)

- TOTAL OPERATIONAL COST = 102 units

- Impact: 44% Cost Reduction! Accepts extra screening tests (FP) to catch almost all critical patients (FN).

High ROC-AUC does not ensure honest probabilities: Random Forest suffers probability compression

Logistic Regression (Naturally Calibrated)

- Reliability Diagram Behavior:
  Traces the ideal 45° diagonal line closely on calibration_curve.

- Probability Meaning:
  When it outputs 80% risk, exactly 80 out of 100 similar patients actually have the disease.

- Business Usability:
  Probabilities are honest and directly actionable for risk scoring and medical decisions.

Random Forest (Probability Compression)

- Averaging Effect:
  Averaging votes across 200 trees pushes probability predictions toward the mean (~0.15 to 0.67).

- Compression Deficit:
  Even for severe patients, it rarely predicts >67% confidence due to tree variance.

- MLOps Engineering Principle:
  Prefer a naturally calibrated architecture over attempting to post-hoc patch flawed outputs.

A strict 4-step sequence prevents data leakage and illusory optimization

STEP 1 — **Select Model Architecture**
Rank candidates via cross-validated ROC-AUC (ranking power).

STEP 2 — **Verify Probability Calibration**
Confirm probabilistic honesty on Reliability Diagram (calibration_curve).

STEP 3 — **Sweep Decision Threshold**
Optimize cutoff τ strictly on Validation fold using the cost matrix.

STEP 4 — **Single-Pass Test Evaluation**
Lock model and threshold τ; evaluate ONCE on holdout Test set.

Reliable decisions require aligning statistical metrics with real business costs

Key Takeaways for AI/ML Systems

1. Stop using default 0.50 threshold blindly:

The 0.50 threshold assumes balanced data and symmetric costs. Always tune threshold τ against a business cost matrix or target recall/precision constraint.

2. Evaluate Discrimination and Calibration separately:

ROC-AUC measures how well a model ranks instances, while Reliability Diagrams measure probability honesty. You need both for high-stakes deployment.

3. Prevent Data Leakage during Threshold Optimization:

Never optimize threshold τ on the test set. Threshold tuning is a hyperparameter decision that must occur strictly within validation folds.

4. Architectural Selection over Post-Hoc Patches:

When probability scores are distorted (e.g., ensemble compression), prefer architectures that are naturally calibrated like Logistic Regression.